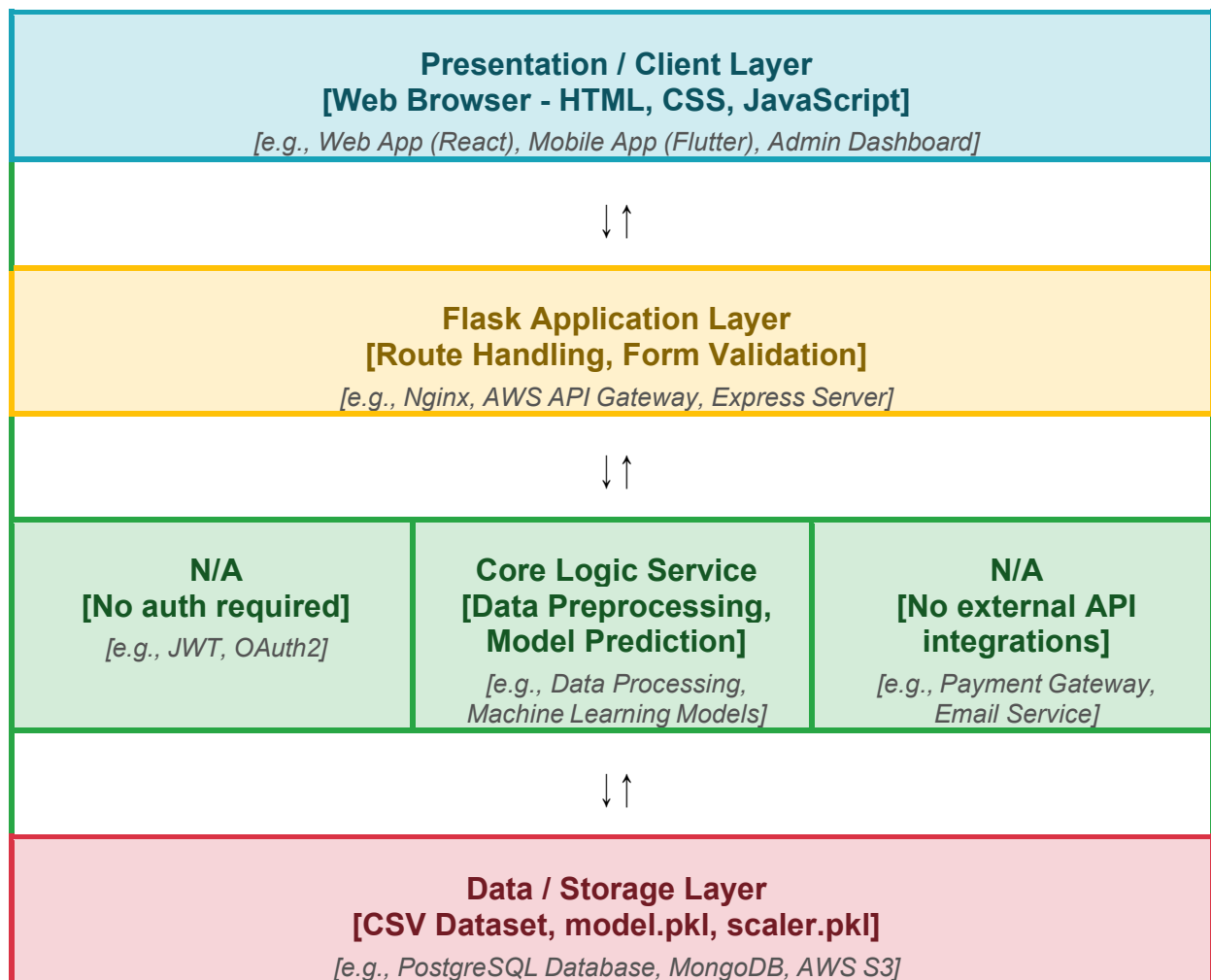


Solution Architecture

Date	15 July 2026
Team ID	SWUID2026217449
Project Name	Rising Waters - Flood Prediction ML
Maximum Marks	5 Marks

Solution Architecture Diagram:

Provide a high-level visual representation of your solution's architecture. Use the structural layout below as a template to map out your Presentation Layer, Application Layer, and Data Layer. Fill in the specific technologies and components for your project directly into the boxes to create your architectural diagram.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
<i>Presentation Layer</i>	<i>Displays the weather input form and prediction result to the user</i>	<i>HTML, CSS, JavaScript</i>
<i>Flask Application Layer</i>	<i>Handles routing, form validation, and calls the ML model for prediction</i>	<i>Flask (Python)</i>
<i>ML Model Layer</i>	<i>Loads trained model and scaler, generates flood risk prediction</i>	<i>Scikit-learn, XGBoost, Pickle</i>
<i>Data Layer</i>	<i>Stores the training dataset and serialized model/scaler files</i>	<i>CSV, Pickle (.pkl)</i>